Sahil Satish Kumar

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Education

NOVA School of Science and Technology — **Lisbon, Portugal**
MSc in Computer Science and Engineering — September 2022 – March 2025
- GPA: **18**/20 (**A** on the ECTS grading system). Thesis grade: **18**/20.
- Relevant coursework: Machine Learning, Deep Learning, Visualization and Data Analytics, High Performance Computing.

NOVA School of Science and Technology — **Lisbon, Portugal**
BSc in Computer Science and Engineering — September 2019 – July 2022
- GPA: **18**/20 (**A** on the ECTS grading system).
- NOVA Young Talent Award 2020-21 (Best student of the first year of the course).
- Relevant coursework: Artificial Intelligence, Probability and Statistics, Databases, Software Engineering.

Experience

Sky Portugal — **Lisbon, Portugal**
Apprentice Backend — July 2025 – August 2025

Project AI Fights Space Debris, Neuraspace — **Lisbon, Portugal**
Scholarship Holder (Master Thesis) — March 2024 – March 2025
- Worked on a Spatio-Temporal Graph Neural Network (STGNN) approach to satellite collision avoidance.
- Collected publicly available TLE data to create a large time-series dataset.
- Built a discrete-time dynamic space conjunction graph based on a transformed evenly spaced over time TLE dataset.
- Achieved 80% test accuracy in conjunction forecasting using RNN/CNN-based STGNNs, outperforming baseline by 10x.

Human-Computer Interaction Institute, Carnegie Mellon University — **Pittsburgh, PA, USA**
Visiting Researcher — October 2024 – December 2024
- Researched various methods and papers in Human-Centered Explainable AI.
- Built a pipeline to convert explanation subgraphs generated by XAI methods into human-readable text using LLMs.
- Designed a user study to evaluate the quality of generated explanations for referral recommendations from GNNs.

NOVA LINCS, NOVA School of Science and Technology — **Lisbon, Portugal**
Undergraduate Researcher — January 2022 – July 2022
- Processed a large dataset from a private healthcare provider to build a doctor referral graph for GNNs.
- Trained a GNN for doctor referral recommendation, achieving 81% test accuracy.
- Developed and implemented a novel causal explanation method for doctor referral recommendations.
- Implemented two baselines for comparison and demonstrated a 4x performance improvement over baseline methods.
- Wrote a research paper about the novel causal explanation method, supporting XAI-driven decision-making in healthcare.

Skills

Programming Languages: Java, Python, C/C++, Kotlin, JavaScript, TypeScript, Dart.

Machine Learning & Data Science: PyTorch, Tensorflow, Scikit-Learn, Pandas, NumPy, Tableau, Spark.

Databases: MySQL, MongoDB, Redis, Neo4j.

Other: Git, Microsoft Azure, Kubernetes, Docker, CUDA, MPI, Linux, Bash, MS Office 365.

Extracurricular Activities

Summer School in Artificial Intelligence and Machine Learning, Lady Margaret Hall — **Oxford, United Kingdom**
Student — August 2022
- Implemented Machine Learning algorithms and Deep Learning models in PyTorch, focused on computer vision.

Scientific Talents in Artificial Intelligence Scholarship, Calouste Gulbenkian Foundation — **Lisbon, Portugal**
Scholarship Holder — September 2021 – July 2022
- Researched various papers about GNNs for recommendations and papers about Causal Inference.
- Developed a causal explanation method for link predictions computed by GNNs.

Languages

Portuguese (Native), **Gujarati** (Native), **Hindi** (Native), **English** (Professional(C1)), **Spanish** (Professional).